



University of Colombo, Sri Lanka



UCSC

University of Colombo School of Computing



**DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**

Academic Year 2025 — 1st Year Examination — Semester 1

IT1306 — Free and Open Source Software for Personal Computing

Multiple Choice Question Paper
(2 Hours)

Important Instructions

- The duration of the paper is **2 Hours**.
- The medium of instructions and questions is English.
- This paper has **40 questions** on **12 pages**. Answer **all** questions.
- All questions are of the **MCQ** (Multiple Choice Questions) type.
- Each question will have **5 (five)** choices with **ONE OR MORE** correct answers.
- This paper consists of 100 marks and all the questions will carry equal marks.
- There will be a penalty for incorrect responses to discourage guessing.
- The mark given for a question will vary from -1 (All the incorrect choices are marked & no correct choices are marked) to +1 (All the correct choices are marked & no incorrect choices are marked). However, **the minimum mark per question would be zero**.
- Answers should be marked on the **special answer sheet** provided.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigilator immediately.
- Mark the correct choices on the question paper first and then transfer them to the given answer sheet which will be machine marked. **Please completely read and follow the instructions given on the other side of the answer sheet before you shade your correct choices.**
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are **not** allowed.
- *All Rights Reserved.* This question paper can NOT be used without proper permission from the University of Colombo School of Computing.

1). Which of the following is/are examples of Free and Open Source Software (FOSS)?

a). PHP	b). Open Office	c). LibreOffice
d). Microsoft Word	e). Adobe Photoshop	

2). Consider the following statements.

- I. An Open Office Application can be installed only on a computer having the UNIX Operating system.
- II. Multiple applications bundled together as a package is referred to as an application suite.
- III. A computer application is a computer program which runs on any operating system.

Which of the above statement(s) is/are correct?

a). (I) Only	b). (II) Only	c). (I) and (II) Only
d). (II) and (III) Only	e). (I) and (III) Only	

Answer questions 3 and 4 based on the following scenario.

You are required to create the following directory structure in the **/home/Gayan** to save a new Writer document named **"Activity01.odt"** inside the directory **"IT1306"**, as shown below:

/home/Gayan → 2026 → IT1306 → "Activity01.odt"

Assumption: You are currently logged in as Gayan and working in **/home/Gayan/** in an Ubuntu server.

3). Which of the following command(s) can be used to create the required directory structure?

a). mkdir 2026-IT1306
b). mkdir /2026/IT1306
c). mkdir IT1306/2026
d). mkdir /IT1306/2026
e). mkdir -p 2026/IT1306

4). Suppose you navigated to directory **"IT1306"**.

Which of the following command(s) will take you to the root directory?

a). cd ..	b). cd /	c). ls ..
d). ls /	e). cd ""	

5). Which of the following command(s) is/are used to remove the files within a directory?

- | | | |
|------------|------------|-----------|
| a). rm | b). rmdir | c). rmDir |
| d). remdir | e). remove | |

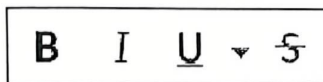
6). Which of the following Word processing applications can be categorized as Free and Open Source Software?

- | | | |
|------------------------|---------------------------|------------------|
| a). Base | b). Word Star | c). Word Perfect |
| d). LibreOffice Writer | e). OpenOffice.org Writer | |

7). Which of the following shortcut key(s) can be used to open the dialog box to find and replace text in LibreOffice Writer?

- | | | |
|--------------|--------------|--------------|
| a). Ctrl + F | b). Ctrl + H | c). Ctrl + R |
| d). Ctrl + G | e). Ctrl + Z | |

8). Which of the following represents the correct functions of the symbols shown in the LibreOffice Writer toolbar?



- | |
|--|
| a). Bold, Italic, Uppercase, Spacing |
| b). Bookmark, Italic, Underline, Style |
| c). Bold, Indent, Underline, Subscript |
| d). Bold, Italic, Underline, Strikethrough |
| e). Bullet, Indent, Underline, Superscript |

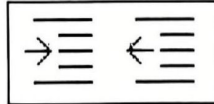
9). Consider the following incomplete sentences.

- I. _____ are portions of a document that appear at the top of every page.
II. _____ appear at the bottom of the page on which they are referenced.
III. _____ gives a more extensive list of alternative words and phrases.

Which of the following options best completes the above incomplete statements?

- | |
|---|
| a). I - Titles, II - Footnotes, III - Glossary |
| b). I - Keywords, II - Captions, III - Dictionary |
| c). I - Footers, II - Endnotes, III - Dictionary |
| d). I - Headers, II - Footnotes, III - Thesaurus |
| e). I - Page Numbers, II - Endnotes, III - Bibliography |

- 10). Identify the functions represented by the toolbar symbols shown below in LibreOffice Writer.



- a). Text Wrap and Anchor
- b). Align Left and Align Right
- c). Increase Indent and Decrease Indent
- d). Line Spacing and Paragraph Spacing
- e). Sort Ascending and Sort Descending

- 11). Which of the following page orientation option(s) is/are available in LibreOffice Writer?

- | | | |
|--------------|---------------|--------------|
| a). Portrait | b). Vertical | c). Inverted |
| d). Diagonal | e). Landscape | |

- 12). What formatting technique(s) in LibreOffice Writer is/are used to format the content in "Paragraph 01" to make it appear as in "Paragraph 02"?

The project VLE consists of course materials, submission links and announcements related to the Software Development Project. Once the students are registered for the project, they will be provided access to the project VLE.

Paragraph 01

The project VLE consists of course materials, submission links and announcements related to the Software Development Project. Once the students are registered for the project, they will be provided access to the project VLE.

Paragraph 02

- | | | |
|-------------------|-------------------|---------------|
| a). Macros | b). Drop Cap | c). Alignment |
| d). Outline Level | e). Text Wrapping | |

- 13). In LibreOffice Writer, what does the Cycle Case option do?

- a). Cycles all letters to uppercase.
- b). Cycles all letters to the opposite case.
- c). Cycles the first letter of every word through uppercase and lowercase only.
- d). Cycles only the first word of a sentence to uppercase (together with proper nouns).
- e). Cycles the selected words through UPPERCASE, lowercase, and Capitalize Every Word.

- 14). Which of the following statement(s) correctly describe(s) features or options in LibreOffice Writer?

- a). The Merge Document option under Track Changes allows combining a document with a data source to create a mail merge.
- b). Endnotes appear at the bottom of the page, while Footnotes are collected at the end of the document.
- c). Authors and reviewers can use the ¶ (Show Formatting Marks) icon to add comments to a document.
- d). Both hyperlinks and cross-references do not update automatically when the linked text changes.
- e). Before inserting a cross-reference to a caption or a list item, the item must be defined as a target using bookmarks or set references.

- 15). Consider the following paragraph.

“A **Bachelor of Information Technology (BIT)** degree provides students with the knowledge and skills to design, develop, and manage *computer systems* and *software applications*. Students learn about programming languages, networking, databases, and ~~cybersecurity~~.”

Which of the following format technique(s) can be seen on the above paragraph?

- | | | |
|-----------------|-------------------|---------------|
| a). Italic | b). Overline | c). Subscript |
| d). Superscript | e). Strikethrough | |

- 16). In a LibreOffice Calc spreadsheet, column A contains first names and column B contains last names. Which formula in column C will combine the first and the last name with a space in between?

- | |
|-------------------------------|
| a). =A1.B1 |
| b). =A1 + B1 |
| c). =A1 * B1 |
| d). =A1 & " " & B1 |
| e). =CONCATENATE(A1; " "; B1) |

- 17). Match the following keyboard shortcut keys in Column A, used in LibreOffice Calc, with their correct cell navigation functions in Column B.

Column A		Column B	
(i)	Ctrl + Page Up	A	Moves the cell focus to the next cell on the right
(ii)	Tab	P	Moves the cell focus up one cell (unless changed by the user)
(iii)	Shift + Tab	C	Moves the cell focus to the same cell on the previous sheet to the left
(iv)	Enter	D	Moves the cell focus down one cell (unless changed by the user)
(v)	Shift + Enter	E	Moves the cell focus to the previous cell on the left

- a. (i) - A, (ii) - C, (iii) - E, (iv) - B, (v) - D
b. (i) - B, (ii) - D, (iii) - A, (iv) - E, (v) - C
c. (i) - E, (ii) - A, (iii) - C, (iv) - D, (v) - B
d. (i) - E, (ii) - C, (iii) - B, (iv) - A, (v) - D
e. (i) - C, (ii) - A, (iii) - E, (iv) - D, (v) - B

- 18). Suppose cell A1 contains the formula $=3+4$ as shown in the following LibreOffice Calc sheet segment.

A1		$\sum$	$=$	$=2+3$
	A	B	C	
1	5			
2				
3				
4				
5				
6				

If the fill handle is used to fill the cells from A2 to A6 based on cell A1, what formula would be pasted in each cell?

- a. A2: $=3+5$; A3: $=3+6$; A4: $=3+7$; A5: $=3+8$; A6: $=3+9$
b. A2: $=3+4$; A3: $=3+4$; A4: $=3+4$; A5: $=3+4$; A6: $=3+4$
c. A2: $=4+3$; A3: $=5+3$; A4: $=6+3$; A5: $=7+3$; A6: $=8+3$
d. A2: $=7+7$; A3: $=8+8$; A4: $=9+9$; A5: $=10+10$; A6: $=11+11$
e. A2: Err:509 ; A3: Err:509 ; A4: Err:509 ; A5: Err:509 ; A6: Err:509

- 19). Which of the following is/are correct cell references in an LibreOffice Calc spreadsheet?

- a. G7 b. 12BQ c. CC88
d. A93A e. 900T

20). Following is a section of LibreOffice Calc datasheet prepared to cater following requirements.
A total of 20 employees (E1–E20) are assigned to different departments (D1–D5) in a company randomly.

- Column A (A2:A21) stores the Employee ID.
- Column B (B2:B21) stores the Department assigned.
- Column D (D3:D7) lists the Departments.
- Column E (E3:E7) calculates the number of employees in each department.

	A	B	C	D	E
1	Employee ID	Department			
2	E1	D2			
3	E2	D3		Department List	Employee Count
4	E3	D1		D1	
5	E4	D5		D2	
6	E5	D2		D3	
7	E6	D1		D4	
8	E7	D4		D5	
9	E8	D3			

Which of the following formulas in cell E3 correctly calculates the number of employees assigned to each Department, assuming the formula will be copied down using fill handle?

- =COUNTIF(B2:B21;D3)
- =COUNTIF(B2:B21;"D3")
- =COUNTIF(B2:B21;"D1")
- =COUNTIF(\$B\$2:\$B\$21;D3)
- =COUNTIF(\$B\$2:\$B\$21;"D1")

Answer the questions 21-23, referring to the following spreadsheet created using LibreOffice Calc.

	A	B
1	ID	Mark
2	S1	45
3	S2	65
4	S3	73
5	S4	35
6	S5	89
7	S6	93
8		
9		

21). If a student tries to use the formula given below in cell B8 what will be the output?

=SUMM(B2:B7)

- 0
- 400
- #REF!
- #NAME?
- #VALUE!

22). Which of the following formula(e) correctly count the number of students scoring more than 75?

- | | |
|-----|-----------------------|
| a). | =IF(B2:B7>75,1,0) |
| b). | =SUMIF(B2:B7,75) |
| c). | =COUNTIFS(B2:B7,75) |
| d). | =COUNT(B2:B7,">75") |
| e). | =COUNTIF(B2:B7,">75") |

23). If a student tries to calculate average mark using the formula given below in cell B9 what will be the output?

=ROUNDDOWN (AVERAGE (B2 : B7))

- | | | | | | |
|-----|------|-----|-------|-----|----|
| a). | 0 | b). | 66 | c). | 67 |
| d). | 66.7 | e). | #NUM! | | |

24). A teacher has recorded students' average marks in column F (F2:F7) in a spreadsheet created using LibreOffice Calc. The grading system is:

- If the average is 90 or above, the grade is A; if 75 or above, the grade is B; if 60 or above, the grade is C; and if below 60, the grade is D.

In column G, the teacher wants to calculate the grade using a single formula in cell G2. Which of the following formula(e) will correctly calculate the grade?

- | | |
|-----|---|
| a). | =CHOOSE(F2,"D","C","B","A") |
| b). | =IF(F2>60,"C",IF(F2>75,"B",IF(F2>90,"A","D"))) |
| c). | =IFS(F2>60,"C",IF(F2>75,"B",IF(F2>90,"A","D"))) |
| d). | =IF(F2>=90,"A",IF(F2>=75,"B",IF(F2>=60,"C","D"))) |
| e). | =IFS(F2>=90,"A",F2>=75,"B",F2>=60,"C",F2<60,"D") |

25). Consider the following statements about LibreOffice Base.

- I. A 'Foreign Key' is a field in one table that refers to the primary key of another table
- II. A 'Primary key' field may contain duplicate values.
- III. Primary key columns are automatically indexed.

Which of the above is/are correct regarding Foreign Key, Primary keys or Indexes?

- | | | | | | |
|-----|-----------------|-----|--------------------|-----|---------------|
| a). | I only | b). | II only | c). | I and II only |
| d). | II and III only | e). | All I, II, and III | | |

- 26). In LibreOffice Calc spreadsheet, a student wants to select the entire row containing the active cell using a keyboard shortcut. Which of the following will correctly select the entire row?

- | | | |
|----------------------|------------------------|-----------------------|
| a). Ctrl + Space bar | b). F8 + Left Arrow | c). Shift + Space bar |
| d). F8 + Right Arrow | e). Shift + Left Arrow | |

- 27). Consider the following statements regarding forms in LibreOffice Base:

- I. A form in Base is a structure that is invisible to the user.
- II. A form in Base provides interaction with the database.
- III. The simplest way to create a form is by using the Form Wizard.

Which of the above statements is/are correct?

- | | | |
|---------------------|------------------------|-------------------|
| a). I only | b). II only | c). I and II only |
| d). II and III only | e). All I, II, and III | |

Answer the questions 28-31, referring to the following Database created using LibreOffice Base.

A college manages its records using LibreOffice Base. The database CollegeDB contains two tables.

- **Students:** StudentID (Integer, Primary Key), StudentName (Text), CourseID (Integer, Foreign Key), Year (Integer)
- **Courses:** CourseID (Integer, Primary Key), CourseName (Text), Department (Text)
- The tables are linked via CourseID, and the administrator uses queries to extract meaningful information.

- 28). Which SQL statement(s) correctly create(s) a foreign key relationship between the *Students* and *Courses* tables in LibreOffice Base?

- | |
|--|
| a). CREATE RELATION Students AND Courses; |
| b). ADD FOREIGN KEY Students.CourseID TO Courses.CourseID; |
| c). ALTER TABLE Students ADD CourseID REFERENCES Courses; |
| d). ALTER TABLE Students ADD FOREIGN KEY (CourseID) REFERENCES Courses(CourseID); |
| e). ALTER TABLE Courses ADD FOREIGN KEY (CourseID) REFERENCES Students(StudentID); |

- 29). Which of the following query/queries correctly display(s) the student names along with their corresponding course names?

a). SELECT StudentName, CourseName FROM Students, Courses;
b). SELECT * FROM Students, Courses WHERE StudentID = CourseID;
c). SELECT StudentName AND CourseName FROM Students WHERE CourseID;
d). SELECT StudentName, CourseName FROM Students JOIN Courses ON Students.CourseID = Courses.CourseID;
e). SELECT StudentName AND CourseName FROM Students WHERE Students.CourseID = Courses.CourseID;

- 30). Which of the following query/queries list(s) students who belong to the 'Science' department?

a). SELECT StudentName FROM Courses WHERE Department = 'Science';
b). SELECT StudentName FROM Students WHERE Department = 'Science';
c). SELECT StudentName FROM Students, Courses WHERE Department = 'Science';
d). SELECT StudentName, CourseName FROM Students ON Courses WHERE Courses.Department = 'Science';
e). SELECT StudentName FROM Students JOIN Courses ON Students.CourseID = Courses.CourseID WHERE Courses.Department = 'Science';

- 31). Which of the following query/queries show(s) the number of students enrolled in each course?

a). SELECT CourseID, COUNT(StudentID) FROM Students;
b). SELECT CourseName, COUNT(StudentID) FROM Courses;
c). SELECT COUNT(StudentID) FROM Students WHERE CourseID;
d). SELECT CourseID, COUNT(StudentID) FROM Students GROUP BY CourseID;
e). SELECT CourseName, COUNT(StudentID) FROM Students GROUP BY Courses;

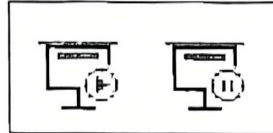
- 32). Which of the following file extensions is the default for LibreOffice Impress presentations?

a). .ppt b). .odp c). .ods
d). .imp e). .docx

33). Which shortcut key is commonly used to start a slide show from the first slide in Impress?

- | | | |
|-----------------|-------------------|---------------|
| a). F2 | b). F5 | c). Ctrl + F1 |
| d). Shift + F12 | e). Shift + Enter | |

34). Which of the following functions are represented by the toolbar symbols shown below in LibreOffice Impress?



- | |
|--|
| a). Start and stop animations. |
| b). Start and stop slide transitions. |
| c). Play the video in a slide and pause the video. |
| d). Play the slide show and pause the slide show. |
| e). Start slide show from the beginning and start slide show from the current slide. |

35). Consider the following statements:

- I. Handout View allows editing notes of a specific slide.
- II. Slide Sorter View allows you to apply transition effects to one or multiple slides.
- III. Outline View can be used to reorder slide titles and modify the text structure of slides.

Which of the above statement(s) is/are correct about the different views in LibreOffice Impress?

- | | | |
|---------------------|------------------------|--------------|
| a). I only | b). II only | c). III only |
| d). II and III only | e). All I, II, and III | |

36). Consider the following incomplete sentence.

“_____ is suitable for a variety of image manipulation tasks, including photo retouching, image composition, and image construction.”

What is/are the appropriate word(s) to complete above incomplete sentence?

- | | | |
|--------------|------------------|-------------|
| a). GIMP | b). MS Publisher | c). Impress |
| d). Inkscape | e). Audacity | |

37). Consider the following image:



Which of the following selection tool(s) is/are used in the above image?

- | | | |
|---------------------|-----------------------|-------------------------|
| a). Free Selection | b). Fuzzy Selection | c). Rectangle Selection |
| d). Select By Color | e). Ellipse Selection | |

38). Which of the following tool(s) in GIMP can be used to improve or adjust colors in an image?

- | | | |
|------------|-------------------|-------------------|
| a). Shear | b). Equalize | c). Color Enhance |
| d). Rotate | e). Gaussian Blur | |

39). GEGI. stands for,

- | |
|-------------------------------------|
| a). Generic Graphics Library |
| b). Graphics Engine for Linux |
| c). GIMP External Graphics Link |
| d). General Editing Graphics Layout |
| e). GIMP Enhanced Graphics Language |

40). Which of the following is/are the native file format of GIMP that preserves layers, channels, a other editing information?

- | | | |
|---------|---------|---------|
| a). XCF | b). JPG | c). GIF |
| d). PNG | e). BMP | |
